

Physics-Informed Deep Stereo: Active Asymmetric Polarization for Data-Efficient Transparent Obstacle Detection

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Abstract—

I. INTRODUCTION

Transparent obstacles represent a fundamental challenge for autonomous navigation and robotic manipulation. Their unique optical characteristics—specifically light transmission, specular reflection, and refraction distortion—render them effectively invisible or geometrically ambiguous to traditional sensing modalities.

Standard obstacle detection methods struggle significantly in this domain. LiDAR sensors fail as emitted pulses transmit through transparent surfaces rather than reflecting back [1]. Conventional RGB-D cameras produce incomplete depth maps with large missing regions (holes) due to strong specular reflections and lack of texture [2]. Although ultrasonic sensors are unaffected by optical transparency, they lack the spatial resolution and directional precision required for dense 3D reconstruction [3].

Recent learning-based methods have shown promise in addressing these limitations. Depth completion approaches train neural networks to infer complete depth maps from corrupted RGB-D inputs [4]. However, these methods are data-intensive, typically requiring tens of thousands of synthetic training images to learn the complex mapping from sparse depth to dense geometry. Active stereo methods [5] leverage structured light patterns to mitigate textureless issues. While these approaches reduce dependency on scene texture, they remain vulnerable to the complex light transport (e.g., strong specular highlights) inherent to transparent materials, still necessitating extensive training data to generalize.

To address these limitations, we propose a Physics-Informed Deep Stereo (PIDS) framework for data-efficient transparent obstacle detection. Unlike symmetric stereo setups, our approach leverages an active asymmetric polarization configuration. We configure one camera to capture the parallel polarization state (preserving specular highlights) and the other to capture the orthogonal state (suppressing highlights). This setup creates a selective photometric discrepancy: background regions remain consistent across views, enabling reliable stereo matching, while transparent objects exhibit significant intensity differences. This physics-based input representation effectively converts the transparency detection problem into a discrepancy learning task. Consequently, our method enables existing stereo backbones to adapt to transparent scenes via data-efficient fine-tuning, eliminating the need for large-scale specialized datasets or complex architectural modifications. To summarize, our main contributions are:

- **A Physics-Informed Sensing Modality:** We propose an active asymmetric polarization setup that converts transparent object detection into a robust photometric discrepancy task.
- **Data-Efficient Framework:** By leveraging physical cues, we demonstrate that standard stereo backbones can be adapted to transparent scenes with minimal real-world fine-tuning, significantly reducing data requirements compared to SOTA methods.
- **Generic Compatibility:** Our approach is model-agnostic, capable of transforming existing stereo matching networks (e.g., RAFT-Stereo) into transparent-aware sensors without architectural changes.

II. RELATED WORK

A. Transparent Object Detection with RGB-D Sensors

Standard RGB-D cameras fail to capture accurate depth for transparent objects due to light transmission and refraction. Depth completion methods aim to recover the corrupted depth structure caused by transparent objects through deep learning. These approaches can be categorized into two main strategies. The first, exemplified by Clear Grasp [4], combines learned predictions (surface normals, masks, occlusion boundaries) with global optimization algorithms to reconstruct depth from over 50,000 synthetic training images. The second strategy, adopted by TransCG [6] and ASGrasp [5], employs end-to-end networks that directly output completed depth maps. The key distinction between TransCG and ASGrasp lies in their input representation: TransCG retains the noisy depth map from RGB-D sensors as input, while ASGrasp discards the unreliable depth measurements entirely and directly utilizes RGB images paired with raw infrared (IR) stereo images to reconstruct geometry. Despite these different strategies, all methods remain data-intensive, requiring tens of thousands of training images to learn the complex mappings from corrupted or incomplete inputs to accurate depth.

B. Polarization-Based Transparent Object Vision

Polarization imaging reveals the geometric properties of transparent surfaces by measuring the polarization state of reflected light (degree of polarization ρ and angle of polarization ϕ), providing physical constraints for 3D reconstruction. Cui et al. [7] explored polarimetric multi-view stereo for 3D modeling in static environments, achieving high-quality reconstruction by resolving polarization ambiguities through graph optimization. However, their multi-view acquisition (6-36 views $\times$ 7 images per view) requires several minutes, rendering it impractical for real-time applications. Zhu & Smith [8] simplified the setup to a stereo pair configuration

(one polarization camera and one RGB camera), jointly solving polarization and stereo constraints to obtain metrically accurate depth maps. While this extends the applicability to broader scenarios, it still necessitates sequential capture of multiple polarization angle images and second-level optimization. Kalra et al. [9] introduced deep learning as an alternative to traditional geometric optimization, employing a monocular multi-polarization camera (simultaneous capture of 4 polarization angles) to achieve real-time segmentation at 20 FPS. However, this approach requires 20,000 labeled training samples and outputs only 2D masks without depth information. All these methods rely on passive illumination, limiting their performance in low-light conditions, and exhibit inherent trade-offs among data efficiency, real-time capability, and 3D perception.

III. METHOD

We present a physics-informed framework for transparent obstacle avoidance that integrates asymmetric polarization cues into a standard stereo vision configuration. By exploiting the differential polarization response of surfaces to generate physically discriminative cues, the system preserves the geometric advantages of stereo while providing more informative inputs for learning. Rather than designing a new architecture from scratch, we treat the stereo network as a reusable backbone and adapt it to the polarization-encoded inputs via data-efficient fine-tuning. This eliminates the need for complex multi-view rigs or large-scale task-specific training datasets.

A. System Overview

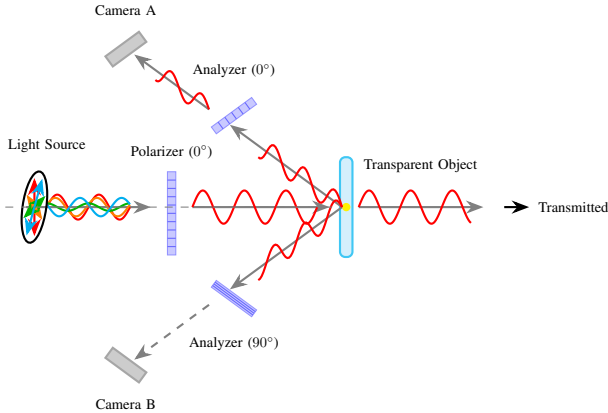


Fig. 1. **Polarization imaging system schematic.** An active light source is polarized at 0° via a linear polarizer. The stereo setup consists of Camera A (Parallel View, $I_{\parallel}$) equipped with a 0° analyzer to capture specular reflections, and Camera B (Cross View, $I_{\perp}$) equipped with a 90° analyzer to suppress them. This asymmetric configuration creates a distinct photometric discrepancy on transparent surfaces while maintaining intensity consistency on diffuse backgrounds.

We employ a physics-based asymmetric optical design to effectively decouple transparent object geometries from complex specular noise at the image-formation level. Specifically, we leverage the polarization-selective properties of linear polarizers. A source-side polarizer ensures that the emitted

illumination maintains a fixed polarization state (0°). Upon interaction with transparent or glossy surfaces, the specular reflection component becomes strongly polarized, whereas ambient and diffuse components remain largely unpolarized.

To exploit this physical behavior, we construct a hybrid stereo configuration. Unlike traditional setups that use identical filters for both views, we introduce an asymmetric filtering scheme: one camera is equipped with an analyzer parallel to the source (0° , Parallel View), while the other is equipped with an analyzer orthogonal to the source (90° , Cross View).

This configuration creates a selective photometric discrepancy. For unpolarized background regions, both cameras capture similar intensities due to diffuse reflection, allowing standard stereo matching algorithms to function correctly. However, for transparent surfaces, the Parallel View captures strong specular highlights, while the Cross View optically suppresses them. This mechanism effectively serves as a hardware-encoded attention signal: the difference between the left and right views explicitly highlights the transparent regions. Consequently, the complex inverse problem of transparency perception is simplified into a differential representation that is easier for neural networks to interpret.

As illustrated in fig. 1, our system integrates a structured-light projector with a source-side polarizer and a calibrated stereo camera pair with distinct polarization states. The outgoing light is pre-polarized, and the returning light is differentially filtered. Crucially, these asymmetric polarization-encoded images serve as a domain-agnostic input representation. By embedding material properties directly into the stereo pair, we ensure that downstream stereo-matching networks can achieve high performance through data-efficient fine-tuning, utilizing the discrepancy cues to localize transparent obstacles.

B. Physics-Informed Input Formulation

We decompose the illumination effects in real-world imaging into several components. Let $I(\mathbf{u})$ denote the measured image intensity at pixel location $\mathbf{u}$. This value is modeled as the linear superposition of three primary factors: diffuse reflection, specular reflection, and background illumination:

$$I(\mathbf{u}) = I_d(\mathbf{u}) + I_s(\mathbf{u}) + I_b(\mathbf{u}) \quad (1)$$

To encode material properties into the stereo pair, we exploit the polarization-dependent nature of the specular term I_s . According to Malus’s law, the intensity of the polarized specular component after filtering varies as $I_s^{filtered} \propto \cos^2 \theta$, where θ is the relative angle between the source polarization and the analyzer.

Our asymmetric setup deliberately creates a photometric discrepancy by configuring the stereo cameras with distinct polarization states: a Parallel View ($I_{\parallel}$, $\theta = 0^\circ$) and a Cross View ($I_{\perp}$, $\theta = 90^\circ$).

For the Parallel View, the specular component is maximized. Assuming the unpolarized components (I_d and I_b) are attenuated by a factor α (transmission coefficient, typically

≈ 0.5) due to the polarizer, the formulation is:

$$I_{\parallel}(\mathbf{u}) \approx \alpha \cdot (I_d(\mathbf{u}) + I_b(\mathbf{u})) + I_s(\mathbf{u}) \quad (2)$$

In contrast, for the Cross View, the analyzer is orthogonal to the source. The polarized specular reflection is effectively eliminated ($I_s \cdot \cos^2(90^\circ) \approx 0$), while the unpolarized components undergo the same attenuation α :

$$I_{\perp}(\mathbf{u}) \approx \alpha \cdot (I_d(\mathbf{u}) + I_b(\mathbf{u})) \quad (3)$$

This formulation reveals the core mechanism of our approach. By subtracting eq. (3) from eq. (2), the discrepancy implies $I_{\parallel} - I_{\perp} \approx I_s(\mathbf{u})$. Consequently, for background regions (where $I_s \approx 0$), the two views exhibit photometric consistency ($I_{\parallel} \approx I_{\perp}$), ensuring the validity of standard stereo matching. However, for transparent surfaces, the localized intensity difference ($I_{\parallel} \gg I_{\perp}$) serves as a robust physical cue for the network to identify obstacles.

C. Standardized Backbone Utilization

To demonstrate the universality of our approach, we adopt a model-agnostic strategy that treats the stereo network as a generic correspondence engine.

Our asymmetric acquisition setup produces inputs that maintain structural compatibility with standard RGB imagery while introducing a deliberate photometric discrepancy on transparent surfaces. Although this discrepancy violates the standard brightness constancy assumption locally, the background regions remain photometrically consistent ($I_{\parallel} \approx I_{\perp}$). This property ensures that the global geometric context is preserved, allowing standard stereo pipelines to establish reliable correspondence. Crucially, the local intensity mismatch ($I_{\parallel} \neq I_{\perp}$) acts as a strong discriminative cue, which Convolutional Neural Networks (CNNs) can learn to associate with the presence of transparent obstacles without requiring architectural modifications.

A stereo neural architecture is required to process our binocular inputs. In this work, we specifically employ RAFT-Stereo [10] as our primary backbone. Unlike traditional cost-volume approaches (e.g., PSMNet), RAFT-Stereo utilizes a recurrent architecture with Gated Recurrent Units (GRUs) to iteratively refine the disparity field.

This choice is particularly strategic for our asymmetric setup. The iterative mechanism allows the network to progressively resolve the photometric discrepancy between the parallel and cross views: early iterations establish global context from the background consistency ($I_{\parallel} \approx I_{\perp}$), while later iterations refine the object boundaries using the high-frequency structural cues provided by our polarization inputs. Furthermore, as a CNN-based architecture, it retains the inductive bias for locality, making it highly data-efficient for our fine-tuning paradigm compared to Transformer-based alternatives. Formally, the network inference is defined as:

$$\hat{\mathbf{D}} = f_{\theta}(\mathbf{X}_{\parallel}, \mathbf{X}_{\perp}) \quad (4)$$

To achieve the claimed data efficiency, we formulate the training as a transfer learning problem. Instead of random initialization, we initialize the network parameters θ_{init} with

weights θ_{pre} pre-trained on a large-scale opaque dataset (e.g., Scene Flow):

$$\theta_{init} \leftarrow \theta_{pre} \quad (5)$$

The optimization objective is then to find the optimal parameters θ^* that minimize the task-specific loss $\mathcal{L}_{task}$ on our small-scale transparent dataset $\mathcal{D}_{trans}$:

$$\theta^* = \arg \min_{\theta} \mathcal{L}_{task}(\mathcal{D}_{trans}; \theta_{init}) \quad (6)$$

Since our physics-informed inputs structurally align with the source domain of θ_{pre} , the optimization starts from a favorable basin of attraction, requiring significantly fewer iterations to converge than training from scratch.

D. Loss Function and Robustness Strategy

$$\mathcal{L} = \sum_{i=1}^N \gamma^{N-i} \|\mathbf{D}_i - \mathbf{D}_{gt}\|_1 \quad (7)$$

As shown in eq. (7), we adopt the standard loss formulation used in RAFT-Stereo, where the total loss $\mathcal{L}$ is computed as a weighted sum of the Manhattan distances between the predicted disparity $\mathbf{D}_i$ at each iteration i and the ground-truth disparity $\mathbf{D}_{gt}$. Compared to the Euclidean distance, the Manhattan distance corresponds to a Mean Absolute Error (MAE) formulation rather than Mean Squared Error (MSE), imposing a linear penalty on outliers rather than a quadratic one. This property is particularly beneficial in real-world transparent-object scenarios, where specular highlights and refraction effects may introduce occasional large residuals. By reducing the sensitivity to such outliers, the model exhibits improved robustness under realistic sensing disturbances.

The exponential weighting term γ^{N-i} controls the relative importance of predictions at different recurrent iterations. This design emphasizes later predictions, which are expected to be more refined, while still providing supervisory signals to earlier iterations. The total loss is obtained by accumulating the weighted losses across all iterations, encouraging the network to optimize the entire iterative refinement process rather than focusing solely on the final output. As a result, the model learns to progressively resolve challenging disparity ambiguities introduced by transparent surfaces.

IV. EXPERIMENTS

A. Experiment Setup

1) **Hardware and Layout:** We employ a stereo imaging system based on Raspberry Pi Global Shutter (GS) cameras (IMX296LQR-C) equipped with 6 mm focal length lenses and an 65 mm stereo baseline. To implement the asymmetric polarization setup, linear polarizing filters are attached to the camera lenses, with the left and right analyzers oriented at 0° and 90° , respectively. Active illumination is provided by a flat-panel LED light source mounted above the stereo rig and covered with a linear polarizing sheet oriented at 0° to define the polarization state of the incident light. Based on Fresnel reflection analysis, the illumination is arranged to

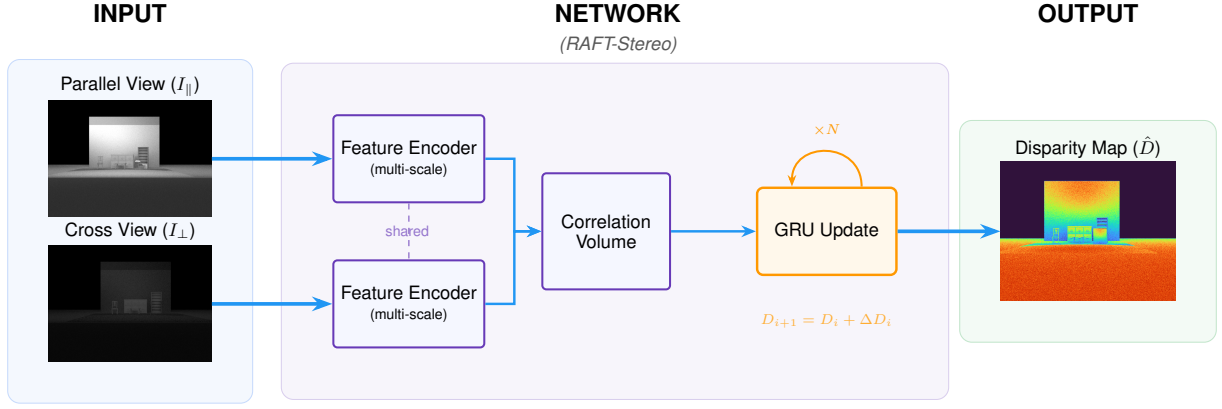


Fig. 2. **Overview of the proposed PIDS framework.** The system takes a pair of active asymmetric polarization images as input: the parallel view ($I_{||}$) captures specular reflections, while the cross view ($I_{\perp}$) filters them out. Features are extracted using weight-sharing Siamese encoders to ensure geometric consistency for background regions. Based on the RAFT-Stereo backbone, the network constructs a correlation volume and employs GRU units for iterative disparity updates ($D_{i+1} = D_i + \Delta D$). This mechanism allows the network to leverage the photometric discrepancy at the input level as a physical cue to accurately refine depth estimates for transparent objects.

favor incidence angles near Brewster’s angle for common glass materials (approximately 56° for $n \approx 1.5$). In practice, this is achieved by tilting the light source by approximately 35° relative to the camera optical axis, which is expected to increase the degree of polarization of specular reflections while maintaining sufficient illumination coverage and field-of-view overlap. Global shutter sensors are employed to avoid rolling shutter artifacts during image acquisition. The hardware configuration is fixed across all experiments to ensure consistency and reproducibility. An auxiliary RGB-D camera is used solely as a reference sensor during data acquisition and does not contribute to the network input.

2) **Radiometric Calibration:** To ensure stable photometric behavior, we apply standard radiometric calibration, including dark-field and flat-field corrections. Dark-field correction removes sensor bias and thermal noise, while flat-field correction compensates for spatially varying sensor gain and optical vignetting. These corrections are applied identically to both stereo views and do not encode any scene-dependent information.

3) **Experimental Environment:** To enable controlled and repeatable evaluation, we conduct experiments in a custom-designed, physically scaled-down chamber with a scale ratio of 1:10. The chamber measures $85 \times 90 \times 75$ cm (width×depth×height) and provides controlled ambient illumination conditions. The interior walls are treated to minimize unintended specular reflections, reducing interference from secondary light bounces. Scene components are primarily fabricated via 3D printing, and all object positions are fixed using concealed base fixtures to ensure repeatability across experiments while preserving geometric configurations representative of real-world scenarios.

B. Training Architecture

We adopt a two-stage fine-tuning training strategy, as illustrated in fig. 3. All data used for fine-tuning are divided into two stages based on their acquisition process, with the majority of the data obtained from simulation. Although

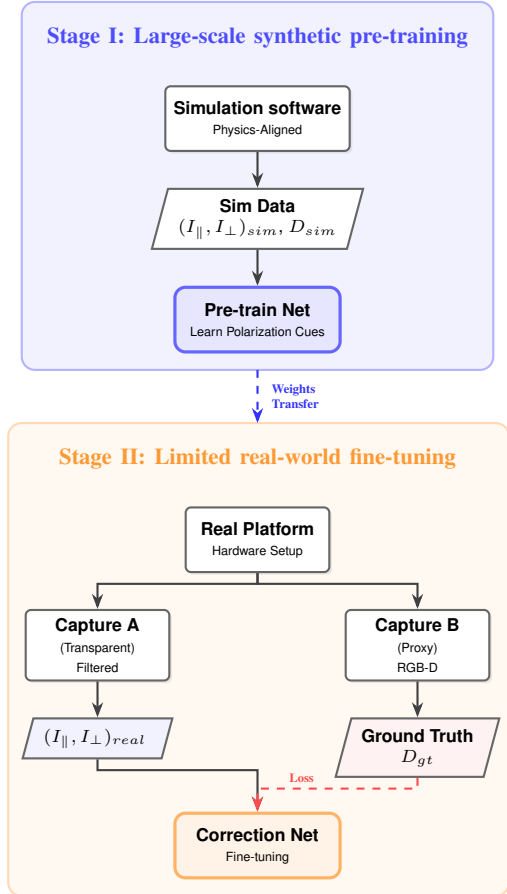


Fig. 3. **Two-Stage Fine-tuning Framework.** Stage I: The stereo network is pre-trained on large-scale physics-aligned synthetic data generated in simulation software to learn polarization-induced photometric cues. Stage II: the same network is fine-tuned on a limited amount of real-world data. Ground-truth depth D_{gt} is obtained using an opaque proxy replacement strategy, where transparent objects are temporarily replaced by opaque counterparts to enable reliable RGB-D sensing. Importantly, the network input always consists of real transparent polarization images, while proxy objects are used solely for supervision.

the training data are generated in a simulated environment, the data generation strictly follows real-world mechanical configurations and sensing geometry to ensure that the fine-tuned network exhibits strong transferability.

After completing training on the simulated data, we further perform a calibration fine-tuning stage using real-world data, while employing the same loss functions to optimize the model. All loss settings are inherited from the model fine-tuned on simulated data to maintain consistency in training objectives across both stages.

To avoid the need for post-processing depth completion when acquiring ground-truth depth maps of transparent objects in real-world scenes, we adopt a proxy object strategy to circumvent this limitation. The proxy objects are fabricated based on one-to-one measurements of the transparent glass objects in the scene and are used solely for spatial localization and volume measurement of the transparent objects to obtain reliable depth supervision. Throughout the entire training pipeline, the proxy objects are not provided as inputs to the neural network.

All fine-tuning stages in this study use stereo image pairs as input and are trained in a supervised manner with corresponding ground-truth depth maps, enabling the network to effectively learn the critical features required to generate depth maps with high completeness.

To support subsequent comparisons with a purely vision-based approach, we additionally fine-tune a network variant with an identical architecture and training protocol, while excluding polarization information from the input. This variant is trained using the same scenes, viewpoints, and amount of training data as the polarization-aware model.

C. Training Data Collection Standards

To ensure high-quality supervision for fine-tuning, we implement a rigorous data curation pipeline designed to eliminate samples that could introduce erroneous geometric or photometric cues. Based on empirical observations, we define the following five quality control criteria:

1) **Geometric Consistency Filtering:** Given that our approach fine-tunes a RAFT-based stereo network, strict geometric rectification is a fundamental prerequisite. We employ feature matching algorithms (e.g., SIFT or ORB) to extract keypoints in the background regions of stereo pairs. Samples exhibiting an average vertical disparity exceeding 1 pixel are identified as rectification failures and are excluded from the training set.

2) **Background Photometric Consistency Filtering:** To verify illumination stability and radiometric calibration, we evaluate the photometric consistency of background regions across polarization channels. Theoretically, diffuse background regions should exhibit negligible polarization effects. Therefore, the absolute intensity difference $|I_{\parallel} - I_{\perp}|$ in these areas is expected to approach zero. Samples displaying significant luminance fluctuations or global exposure shifts are discarded.

3) **Polarization Signal Validity Filtering:** The efficacy of the proposed method relies on the polarization intensity

difference observed on transparent surfaces. We inspect the specular highlight regions of transparent objects to ensure $I_{\parallel} > I_{\perp}$. Samples where the polarization-induced contrast is insufficient (i.e., $I_{\parallel} \approx I_{\perp}$), potentially due to suboptimal incident angles or polarization misalignment, are removed to prevent feature dilution.

4) **Ground Truth Alignment Filtering:** Precise alignment between input images and ground-truth depth is critical, particularly given the manual proxy-swapping procedure. We compute the structural consistency of background edges between the left input image ($I_{\parallel}$) and the RGB reference from the depth sensor. Instances exhibiting misalignment artifacts or pixel displacement greater than 1 pixel are rejected to avoid learning incorrect depth boundaries.

5) **Depth Validity Rate Filtering:** To guarantee meaningful supervision signals, we impose a completeness constraint on the ground-truth data. We require the valid depth pixel ratio within the transparent object mask to exceed 90%. Proxy depth maps suffering from sparsity or excessive sensor noise in the region of interest are excluded.

D. Training Configuration

1) **Stage I:** In Stage I, the objective is to pre-train the network to internalize polarization-induced geometric features using high-fidelity synthetic data. We employ the AdamW optimizer with an adjusted ϵ value set to 10^{-6} to ensure numerical stability. This adjustment mitigates gradient instability caused by extreme intensity gradients in specular highlights, which may exceed the numerical tolerance of the default configuration. To accelerate convergence, a OneCycle learning rate policy is adopted. Given the high quality of the synthetic dataset, a relatively large batch size is used to provide more accurate gradient estimates, promoting stable optimization. To limit the overall training budget, the number of training iterations is reduced and the evaluation frequency is increased to capture the optimal convergence point of the loss curve.

2) **Stage II:** Stage II focuses on domain adaptation using a limited amount of real-world data. We continue to utilize AdamW with a significantly reduced learning rate to prevent catastrophic forgetting of the physics-based priors acquired in Stage I. This stage aims to calibrate the network to real-world sensor noise and environmental variations while preserving the learned polarization cues. To evaluate data efficiency, the total number of fine-tuning iterations is strictly constrained, encouraging rapid adaptation without overfitting to sample-specific noise.

E. Quantitative Standards

As the primary objective of this study is to optimize depth map completeness and data utilization efficiency, we define quantitative evaluation metrics along two dimensions. For depth estimation accuracy, we adopt the End-Point Error (EPE) and the Root Mean Squared Error (RMSE). EPE measures the average absolute difference between the predicted and ground-truth disparity in pixels. RMSE is employed to evaluate depth errors and is more sensitive to large

deviations and outliers, making it particularly suitable for assessing whether severe distortions caused by specular reflections are mitigated. To prevent background structures from diluting the evaluation, all metrics are computed exclusively within the transparent object masks. To further evaluate data utilization efficiency, we analyze the EPE performance of models trained with varying amounts of training data (e.g., 1%, 10%, and 100%), enabling a direct assessment of how effectively each method leverages available supervision.

F. Comparison with Baseline RAFT-Stereo

G. Controlled Micro-Scale Depth Completeness Evaluation

H. Real-World Robot Evaluation

V. RESULTS

VI. CONCLUSIONS

APPENDIX

ACKNOWLEDGMENT

Portions of this manuscript were refined for clarity and language, and limited code editing support was provided using AI-based assistance tools, specifically Google Gemini and Grammarly. The authors are fully responsible for the research ideas, methodology, implementation, experimental results, and conclusions presented in this work.

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